Climate Change: Global Status Report 2024

This report summarizes the current state of climate change, key indicators, international agreements, and projected impacts on ecosystems and human societies.

1. Current Global Temperature Trends

Global average surface temperatures have risen by approximately 1.2 degrees Celsius above pre-industrial levels as of 2024. The last decade (2014-2024) has been the warmest on record. The Intergovernmental Panel on Climate Change (IPCC) warns that exceeding 1.5 degrees Celsius will trigger more severe and irreversible consequences.

The Arctic is warming at more than twice the global average rate — a phenomenon known as Arctic amplification. Sea ice extent in September 2023 reached a record low of 4.23 million square kilometers, well below the 1981-2010 average of 6.3 million square kilometers.

1.1 Temperature Anomalies by Region

Region	Warming (°C)	Trend	Risk Level
Arctic	+3.1	Accelerating	Critical
Sub-Saharan Africa	+1.8	Steady	High
South Asia	+1.5	Steady	High
Europe	+1.9	Accelerating	High
North America	+1.6	Steady	Medium-High
Oceania	+1.4	Steady	Medium

2. Greenhouse Gas Emissions

Carbon dioxide (CO_2) concentrations reached 421 parts per million (ppm) in 2023, the highest level in at least 800,000 years. Methane (CH_4) concentrations are now more than 2.5 times pre-industrial levels, driven by agriculture, fossil fuels, and wetland emissions.

The energy sector remains the largest source of greenhouse gas emissions globally, accounting for approximately 73% of total emissions. Land-use change, deforestation, and agriculture together contribute around 18.4% of global emissions.

2.1 Top Emitting Countries (2023)

Country	CO_2 (Gt/yr)	% of Global	Per Capita (t)
China	11.47	30.9%	8.0
United States	5.01	13.5%	14.9

India	2.83	7.6%	2.0
Russia	1.83	4.9%	12.6
Japan	1.06	2.9%	8.5
Germany	0.67	1.8%	8.0
South Korea	0.61	1.6%	11.9

3. Paris Agreement and Policy Response

The Paris Agreement, adopted in 2015 and entered into force in 2016, represents the most comprehensive international climate framework to date. Under the agreement, 196 parties have committed to Nationally Determined Contributions (NDCs) to limit warming to well below 2 degrees Celsius, with efforts to limit it to 1.5 degrees.

As of 2024, current NDC pledges put the world on track for approximately 2.7 degrees of warming by 2100. The UN Environment Programme's Emissions Gap Report consistently finds that pledged actions are insufficient to meet Paris Agreement goals.

3.1 Key Milestones

2015 **-** Paris Agreement adopted at COP21 in Paris, France.

2019 **-** European Union announces European Green Deal targeting climate neutrality by 2050.

2021 **-** COP26 in Glasgow: countries pledged to phase down coal and end deforestation by 2030.

2022 **-** US Inflation Reduction Act: $369 billion in climate and clean energy investments.

2023 **-** COP28 in Dubai: first Global Stocktake shows insufficient progress toward Paris goals.

2024 **-** Over 130 countries have adopted net-zero targets covering 88% of global emissions.

4. Projected Impacts

Sea level rise is projected to reach 0.3 to 1.0 meters by 2100 under various emissions scenarios. Approximately 1 billion people live in low-elevation coastal zones at risk of flooding. Miami, Shanghai, Mumbai, and Jakarta are among the most vulnerable cities.

Food security is a major concern. Crop yields for staples like wheat, maize, and rice are projected to decline by 2-6% per decade due to higher temperatures and changing rainfall patterns. The number of people at risk of hunger could increase by 8-80 million by 2050.

Biodiversity loss is accelerating. Climate change is the second largest driver of species extinction after habitat destruction. Coral reefs, which support 25% of marine species, face near-total collapse if warming exceeds 2 degrees Celsius.

5. Solutions and Mitigation Pathways

Renewable energy capacity has grown exponentially. Solar power costs have fallen 89% since 2010, making it the cheapest source of electricity in history. Wind power costs have fallen 70% over the same period. In 2023, renewables accounted for 30% of global electricity.

Carbon capture and storage (CCS) technologies are being scaled. Direct air capture (DAC) facilities can remove CO_2 directly from the atmosphere, though current costs of $400-1000 per tonne of CO_2 remain prohibitively high for large-scale deployment.

Nature-based solutions including reforestation, wetland restoration, and sustainable agriculture could provide up to 30% of the mitigation needed by 2030. The Bonn Challenge aims to restore 350 million hectares of degraded land by 2030.